

[27]

SARDAR PATEL UNIVERSITY
BCA (SEMESTER-I) EXAMINATION 2023
07 March, 2023, Tuesday
02:00PM to 05:00PM
US01FBCA02 Mathematics I



Total Marks:70

Q-1 Select the correct option for each of the followings : [10]

- 1 A function $f: A \rightarrow B$ is said to be _____ if each element of B is the image of some element of A
 (a) onto (b) one to one (c) invertible (d) none
- 2 which of the following is not group under addition :
 (a) $\mathbb{N}$ (b) $\mathbb{Z}$ (c) $\mathbb{Q}$ (d) $\mathbb{R}$
- 3 If $U = \{1, 2, 3, 4, 5\}$, $A = \{1, 2, 3\}$, $B = \{4, 5\}$ then $(A/B) \cap U =$ _____
 (a) $\emptyset$ (b) U (c) $\{1, 2, 3\}$ (d) $\{4, 5\}$
- 4 The number of elements in power set of empty set is _____
 (a) 1 (b) 2 (c) 3 (d) 4
- 5 A square matrix A is said to be skew symmetric if _____
 (a) $A + A^T$ (b) $A = -A^T$ (c) $A = A^T$ (d) none
- 6 $D = \text{diag}(d_{ij})$ is diagonal if its _____ elements are all zero .
 (a) diagonal (b) non diagonal (c) all (d) none
- 7 An element e is said to be left identity element for binary operation $*$ if
 (a) $a * e = e$ (b) $e * a = e$ (c) $a^{-1} * a = e$ (d) $a * a^{-1} = e$
- 8 The mean of first 10 natural numbers is
 (a) 5 (b) 5.5 (c) 6.5 (d) 0
- 9 The geometric mean for the frequency distribution is given by _____
 (a) $\text{Antilog}(\sum f \log x / n)$ (b) $\sum f \log x / N$ (c) $\sum f / x$ (d) none
- 10 The observation whose frequency is highest is called _____
 (a) mean (b) median (c) mode (d) none

Q-2 Do as Directed (any 10) [20]

- 1 Write down power set of $\{1, 3, 6\}$
 - 2 State the principle of mathematical induction in two equivalent forms
 - 3 For rational number a and b Define $a * b = ab/2$. Find identity element for given binary operation $*$.
 - 4 Let $B = \{1, 2\}$. Is B closed under
 (a) multiplication (b) addition
 - 5 If $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{4, 5, 6, 7, 8, 9\}$ and $C = \{2, 3, 5, 7, 8, 9\}$,
 Find $(A \cap B) \oplus (B \cap C)$
 - 6 Show that right cancellation law holds in group G .
 - 7 Find $\|u\|$ if $u = (3, -12, -4)$
 - 8 Find x if $B = \begin{pmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{pmatrix}$ is skew symmetric
 - 9 Find Mean.
- | | | | | | | | | |
|---|----|----|----|----|----|----|----|----|
| X | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| F | 14 | 21 | 25 | 43 | 51 | 40 | 39 | 12 |
- 10 Find the Geometric mean of 2, 4, 8, 12, 16.
 - 11 Prove that with the help of Algebra of sets : $A \cap (A \cup B) = A$
 - 12 If A and B are symmetric then, Is it true that AB symmetric ?

(P.T.O.)

Q-3 A Prove that $P(n): 1+4+7+\dots+(3n-2) = n(3n-1)/2$ [05]

B Prove that with the help of laws of Algebra of sets:
(i) $(\emptyset \cup A) \cap (B \cup A) = A$ (ii) $(B^c \cap U) \cap (A^c \cup \emptyset) = (A \cup B)^c$ [05]

OR

Q-3 A If Q is defined by $Q(a,b) = \begin{cases} 0 & \text{if } a < b \\ Q(a-b, b) + 1 & \text{if } a \geq b \end{cases}$ [05]

Then find $Q(2,3)$, $Q(19,5)$

B For sets A and B prove that,
 $A \oplus B = (A/B) \cup (B/A)$ [05]

Q-4 A Let $f(x) = 2x^2 + 4x + 4$ and $g(x) = 3x^3 + 6x + 3$. Find roots of $f(x)$ over Z_{10} and Find roots of $g(x)$ over Z_3 [05]

B Define : (i) commutative ring (ii) unit element in a ring
(iii) $(a^{-1})^{-1} = a$, for any element a in G . [05]

OR

Q-4 A Define $*$ on Q by $a*b = a+b-ab$. [05]

(i) Is $(Q, *)$ semigroup?

(ii) Is $(Q, *)$ monoid?

B Consider the set Z = the set of integers. Which of the following operations in Z are associative: (i) addition (ii) multiplication (iii) subtraction. Justify your answers [05]

Q-5 A If $A = \begin{bmatrix} 1 & 6 & 2 \\ 3 & -1 & 3 \\ 3 & 1 & 5 \end{bmatrix}$ then find AA^T [05]

B Solve by determinants : $\begin{cases} 2x - 4y = 7 \\ 3x - 6y = 5 \end{cases}$ [05]

OR

Q-5 A Let $u = (5, 4, 1)$, $v = (-3, 0, 4)$ and $w = (0, 5, -8)$ and $k = 2$ find (i) $5u + 2v + 3w$
(ii) $\|w\|$ (iii) $(k \cdot v) + w$ (iv) which of the two vectors are orthogonal

B If $A = \begin{bmatrix} 1 & 4 & 5 \\ 1 & 3 & 6 \\ 1 & 5 & 7 \end{bmatrix}$ $B = \begin{bmatrix} 2 & -2 & 1 \\ 0 & 4 & 5 \\ 2 & 6 & 4 \end{bmatrix}$ then find $3A + 2B$ [05]

Q-6A For the given data harmonic mean and geometric mean. [05]

x_i	10	20	30	40	50	60	70
f_i	7	20	43	58	35	18	4

- B Calculate Median and Mode for the following data. [05]

Weight (lbs) X	130	135	140	145	146	148	149	150	157
No. of persons	3	4	6	6	3	55	2	1	1

OR

- Q-6A From the following data calculate the missing value when Mean is 115.86 [05]

X	110	112	113	117	?	125	128	130
f	25	17	13	15	14	8	6	2

[05]

- B The following table shows the wage distribution of the workers.

Wages in Rs.	No. of workers
100-300	35
300-500	70
500-700	105
700-900	210
900-1100	65
1100-1300	20
1300-1500	08

Calculate the median wage of workers.

— X —

